WEEK 4

1. $\{(-\infty, r] : r \in \mathbb{Q}\}$

$\mathcal{B}_{\mathbb{R}} = \sigma(\{\text{set of all right open left closed intervals}\})$

$= \sigma(\{(a,b] : a, b \in \mathbb{R}\})$

suppose, $\mathcal{C} = \{(a,b] : a, b \in \mathbb{R}\}$

$\mathcal{G} : \{(-\infty, r] : r \in \mathbb{Q}\}$ is

trivially $(-\infty, r] \in \mathcal{C}$.

$\Rightarrow \mathcal{G} \subseteq \mathcal{C} \subseteq \sigma(\mathcal{C})$

$\mathcal{G} \subseteq \sigma(\mathcal{C})$

Now consider $(a,b]$ s.t. $a, b \in \mathbb{Q}$

then

$(a,b] = (-\infty, b] \cap ((-\infty, a])^c$

$\in \sigma(\mathcal{G})$ $\in \sigma(\mathcal{G})$

$\in \sigma(\mathcal{G})$

$\Rightarrow [a,b] \in \sigma(\mathcal{G})$

consider $(a,b]$ s.t. $a \in \mathbb{R}$ & $b \in \mathbb{Q}$

$[a,b] = \bigcap_{n=1}^{\infty} (a_n, b]$ $\in \sigma(\mathcal{G})$

a_n : n^{th} digit decimal rep. of a.

$[a,b] \in \sigma(\mathcal{G})$ ∵ it's countable intersection of sets from $\sigma(\mathcal{G})$

$(a,b] = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b]$

$\sigma(\mathcal{G})$

countable union of sets of $\sigma(\mathcal{G})$

Now, $(a,b]$ where $a \in \mathbb{Q}$, $b \in \mathbb{R}$.

then $(a,b) = \underbrace{\bigcup_{n=1}^{\infty} (a, b_n]}_{\in \sigma(\mathcal{G})}$ $\quad b_n = n^{th}$ digit decimal rep. of b

Also, $(a,b] = \underbrace{\bigcap_{n=1}^{\infty} (a, b+\frac{1}{n})}_{\in \sigma(\mathcal{G})}$

Also,

if $(a,b]$ where $a \in \mathbb{R}$ & $b \in \mathbb{R}$

$(a,b) = \bigcup_{n=1}^{\infty} (a, b_n]$ $\quad b_n = n^{th}$ digit decimal rep.

Part 2: & $(a,b] = \bigcap_{n=1}^{\infty} (a, b+\frac{1}{n})$

Now, consider $\mathcal{T}_1 = \{\{x\} : x \in \mathbb{R}\}$

& $\mathcal{T}_2 = \{A \subseteq \mathbb{R} : A \text{ is finite}\}$

$\mathcal{T}_3 = \{E \subseteq \mathbb{R} : E \text{ is countable or } E^c \text{ is countable}\}$

Now, claim: $\sigma(\mathcal{T}_1) = \mathcal{T}_3$ ← This can be easily proved

$\mathcal{T}_1 \subseteq \mathcal{T}_3$ (∵ set of all singletons is countable also $\mathcal{T}_3$ is a σ-field)

Now, any $A \in \mathcal{T}_3 \Rightarrow A$ is countable of A^c countable

Case I: consider A is countable i.e. there is bijection on A from $\mathbb{N}$

then, $A = \{a_1, a_2, \dots\}$

$A = \underbrace{\bigcup_{i=1}^{\infty} \{a_i\}}_{\in \sigma(\mathcal{T}_1)} \quad \in \sigma(\mathcal{T}_1)$

Case II: A^c is countable

$A^c = \bigcup_{i=1}^{\infty} \{a_i\} \quad \in \sigma(\mathcal{T}_1)$

then $A = (A^c)^c \in \sigma(\mathcal{T}_1)$

hence $\mathcal{T}_3 \subseteq \sigma(\mathcal{T}_1) \Rightarrow \sigma(\mathcal{T}_1) = \mathcal{T}_3$

Now, consider set $[0,1]$ Now, $[0,1] \notin \mathcal{T}_3$

∵ Neither $[0,1]$ is countable nor $[0,1]^c$.

Hence, ∃ element $[0,1] \in \mathcal{B}_{\mathbb{R}}$ & $[0,1] \notin \mathcal{T}_3$

hence $\sigma(\mathcal{T}_1) = \mathcal{T}_3 \neq \mathcal{B}_{\mathbb{R}}$

Also, $\sigma(\tau_2) = \tau_2$

2. $\mathcal{C}$ is monotone class of $\mathbb{R}^d$ containing all open & closed sets.

To Prove: $\mathcal{B}_{\mathbb{R}^d} \subseteq \mathcal{C}$

Now, consider $\mathcal{F} = \left\{ \bigcup_{i=1}^{n} (a_i, b_i] \;\middle|\; a_i, b_i \in \mathbb{R}^d \text{ & } n \in \mathbb{N} \right\}$ s.t. all $(a_i, b_i]$'s are disjoint

$A \in \mathcal{F}$ $A =$

We already know $\mathcal{F}$ is a field.

Now, and $A \in \mathcal{F} \Rightarrow A = \bigcup_{i=1}^{n} (a_i, b_i]$ for some $a_i, b_i \in \mathbb{R}$ & $n \in \mathbb{N}$

Now, $(a_i, b_i] = \bigcap_{n=1}^{\infty} (a_i, b_i + \frac{1}{n})$

Now, $(a_i, b_i + \frac{1}{n})$ is an open interval & hence open set. i.e. $(a_i, b_i + \frac{1}{n}) \in \mathcal{C}$

Also, $\because$ $\mathcal{C}$ is monotone class $\bigcap_{n=1}^{\infty} (a_i, b_i + \frac{1}{n}) \in \mathcal{C}$

$\Rightarrow$ $(a_i, b_i] \in \mathcal{C}$

Now every $(a_i, b_i] \in \mathcal{C}$ for any $a_i, b_i \in \mathbb{R}$

Now consider $S_k = \bigcup_{i=1}^{k} (a_i, b_i]$

certainly $S_k \subseteq S_{k+1}$ hence

Does closure under countable union of increasing sets $\Rightarrow$ closure under finite union of increasing sets

$A_1 \subseteq A_2 = A_2 = A_2 \ldots$

3. (i) $\left(\liminf_{n\to\infty} A_n\right)^c = \limsup A_n^c$

$$\liminf_{n\to\infty} A_n = \bigcup_{k=1}^{\infty}\bigcap_{n=k}^{\infty} A_n$$

$$\left(\liminf_{n\to\infty} A_n\right)^c = \left(\bigcup_{k=1}^{\infty}\bigcap_{n=k}^{\infty} A_n\right)^c = \bigcap_{k=1}^{\infty}\left(\bigcap_{n=k}^{\infty} A_n\right)^c$$

$$= \left(\bigcap_{k=1}^{\infty}\bigcup_{n=k}^{\infty} A_n^c\right)$$

$$= \limsup_{n\to\infty} A_n^c$$

(ii) $\left(\limsup_{n\to\infty} A_n\right)^c = \liminf_{n\to\infty} A_n^c$

$$\limsup A_n = \bigcup_{k=1}^{\infty}\bigcap_{n=k}^{\infty} A_n$$

Same as above

(iii) $\liminf_{n\to\infty} A_n \subseteq \limsup A_n$

Consider $x \in \liminf_{n\to\infty} A_n \Rightarrow x \in A_k \quad \forall k \geq n_0$

$\Rightarrow x \in A_k$ for infinitely many A_k's

Hence $x \in \limsup_{n\to\infty} A_n$

(iv) $A_n \uparrow A$ & $A_n \downarrow A$ then

$$\liminf_{n\to\infty} A_n = \bigcup_{k=1}^{\infty}\bigcap_{n=k}^{\infty} A_k$$

WLOG. $A_n \uparrow A$ So $A_1 \subseteq A_2 \subseteq \ldots$ A & $\bigcup_{n=1}^{\infty} A_n = A$ $\}$ -(*)

Hence. $\bigcap_{n=k}^{\infty} A_k = A_k$

$$\liminf_{n\to\infty} A_n = \bigcup_{k=1}^{\infty} A_k = A$$ Using (*)

$$\bigcup_{n=k}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n = A \qquad \forall k \in \mathbb{N}$$

hence. $\bigcap_{k=1}^{\infty}\left(\bigcup_{n=k}^{\infty} A_n\right) = \bigcap_{k=1}^{\infty}\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} A_n = A$

hence $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$

Now for $A_n \to A \Rightarrow A_n^c \to A^c$

So $\liminf_{n\to\infty} A_n = \left(\liminf_{n\to\infty} A_n\right)^{cc} = \left(\limsup_{n\to\infty} A_n^c\right)^c$

$= \left(A^c\right)^c \quad A$

Ally $\limsup_{n\to\infty} A_n = \left(\left(\limsup_{n\to\infty} A_n\right)^c\right)^c = \left(\liminf_{n\to\infty} A_n^c\right)^c$

$= \left(A^c\right)^c = A$

Q. $1_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{o.w.} \end{cases}$ $A \subseteq \Omega$

$\{A_n\}_n$ be sequence of subsets of Ω.

To Prove: $\liminf_{n\to\infty} 1_{A_n}(\omega) = 1_{\liminf_{n\to\infty} A_n}(\omega)$

$f^{-1}(A) \in \mathcal{B}_{\mathbb{R}^m}$ for $\forall A \in \mathcal{B}_{\mathbb{R}^n}$ open sets

$h(\omega) = f(\omega) 1_A + g(\omega) 1_{A^c}$

consider $B \in \mathcal{B}_{\mathbb{R}}$ $h^{-1}(B)$

all. $h(\omega) \in B$

$f^{-1}(R_f \cap B)$

if $R_f \cap B = \phi$

ϕ

$R_f \cap B$

$h(\omega) \in R_f$ & $h(\omega) \in B$